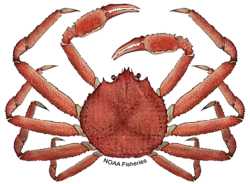
Short-term mechanisms, long-term consequences:
snow crab acclimation to ocean acidification



Alaska Marine Science Symposium
January 2026, Anchorage, AK

Chris Long, NOAA AFSC Kodiak
Laura Spencer, University of Washington SAFS
Jennifer Gardner, NOAA AFSC Kodiak
Ingrid Spies, NOAA AFSC Seattle
Steven Roberts, University of Washington SAFS

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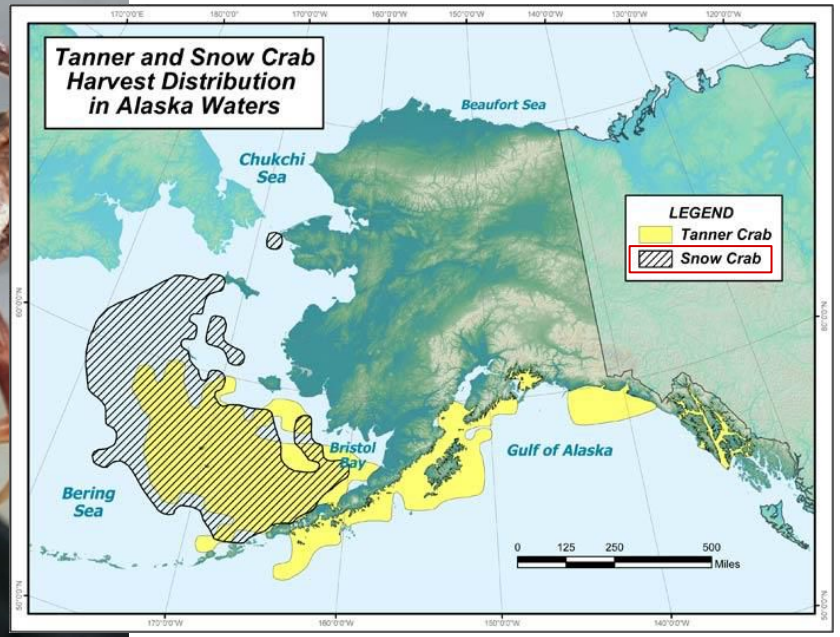
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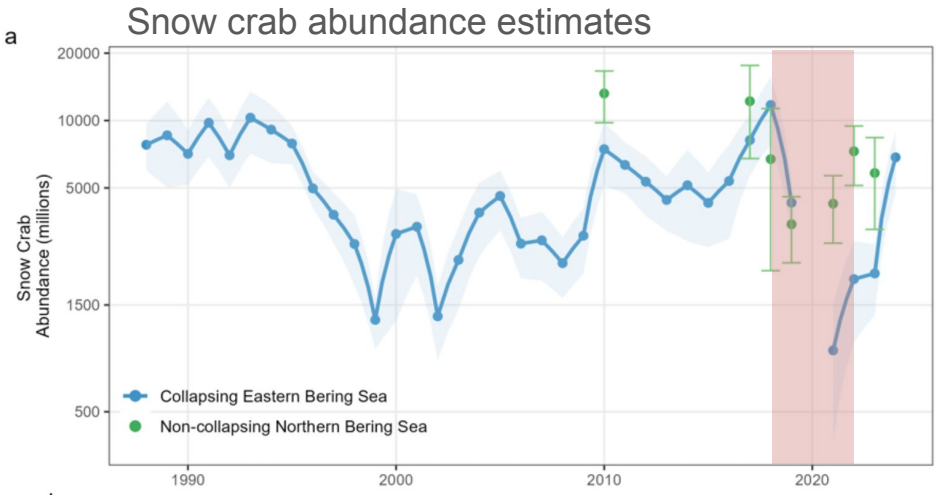
Snow crab, *Chionoecetes opilio*





ADFG

Snow crab population changes linked to environmental conditions



Fedewa, Copeman, & Litzow 2025

Energetic limitations and mass mortality of Bering Sea snow crab: Interacting effects of warming and density on collapse and recovery

Authors: Erin J. Fedewa, Louise A. Copeman, and Michael A. Litzow | AUTHORS INFO & AFFILIATIONS

Publication: Canadian Journal of Fisheries and Aquatic Sciences • 24 November 2025 • https://doi.org/10.1139/cjfas-2025-0099

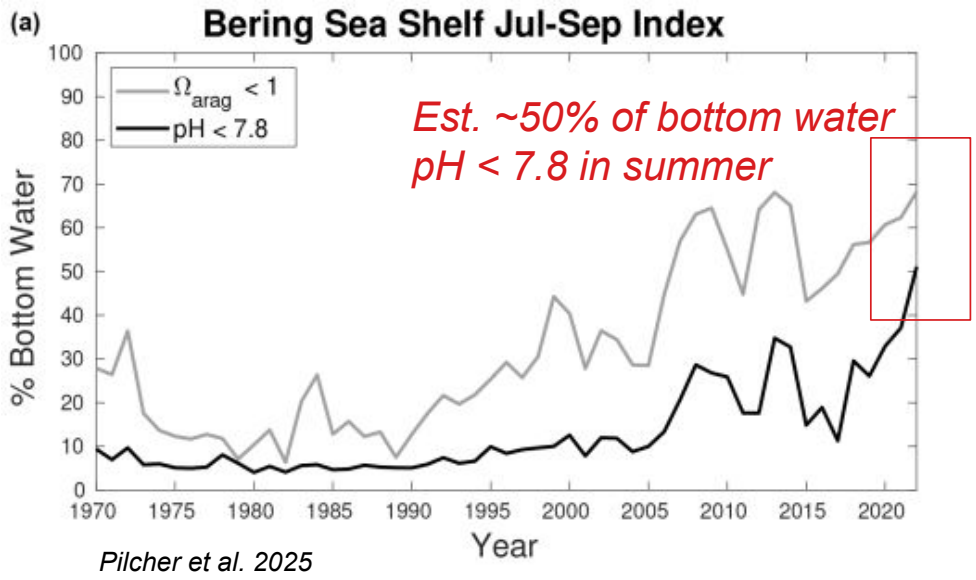
EBS snow crab mass mortality during heatwaves, likely due to starvation, first EBS snow crab fishery closure in history

Recruitment is key to recovery!

Early life stages sensitive to environmental stressors

Does ocean acidification affect recruitment?

Ocean Acidification (OA) is happening in Bering Sea



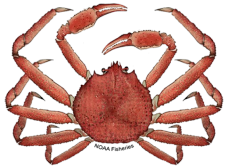
Pilcher et al. 2025

Amplified bottom water acidification rates on the Bering Sea shelf from 1970–2022

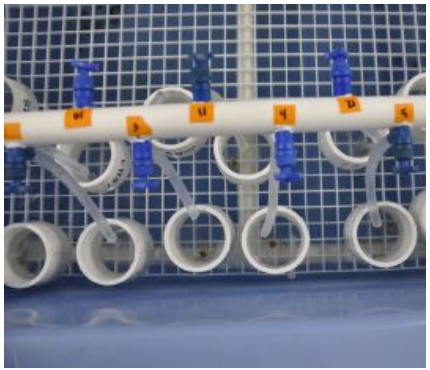
Darren J. Pilcher, Jessica N. Cross, Natalie Monacci, Linquan Mu, Kelly A. Kearney, Albert J. Hermann, and Wei Cheng

NOAA Kodiak Research Station





Today: What are the impacts of OA on snow crab juveniles?



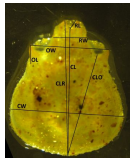


Laboratory Experiment

- Wild juveniles collected during NOAA 2021 EBS survey
- Held for 396 days, 3 pH treatments (at 4°C) in Kodiak lab

Data:

1. Survival, molting, growth, morphometrics
2. Gene expression (RNASeq), 8hr & 88 days

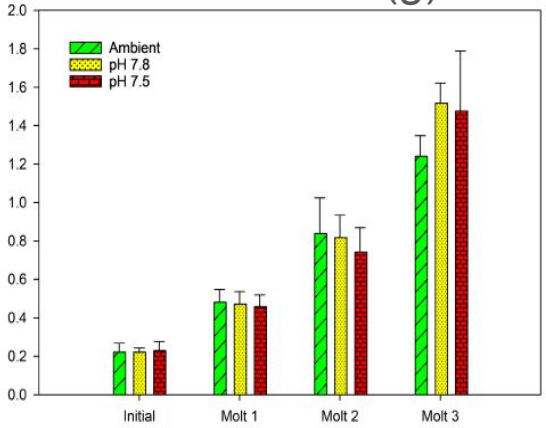


n=30 / treatment

Control	Moderate OA	Severe OA
pH 8.0	*pH 7.8*	*pH 7.5*

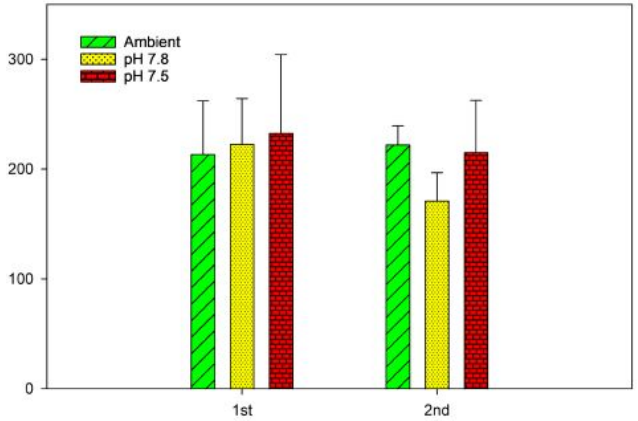
Snow crab juveniles are surprising tolerant of long-term OA

Wet mass (g)



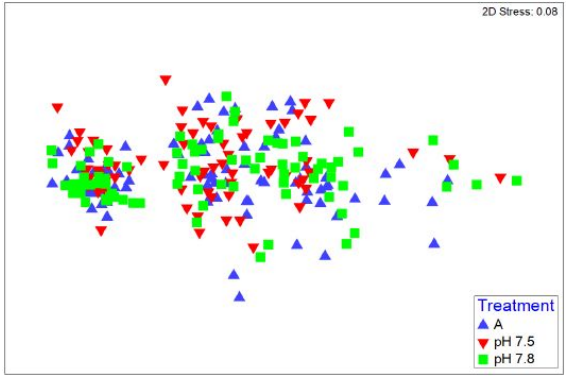
Molt stage

Intermolt duration



Intermolt

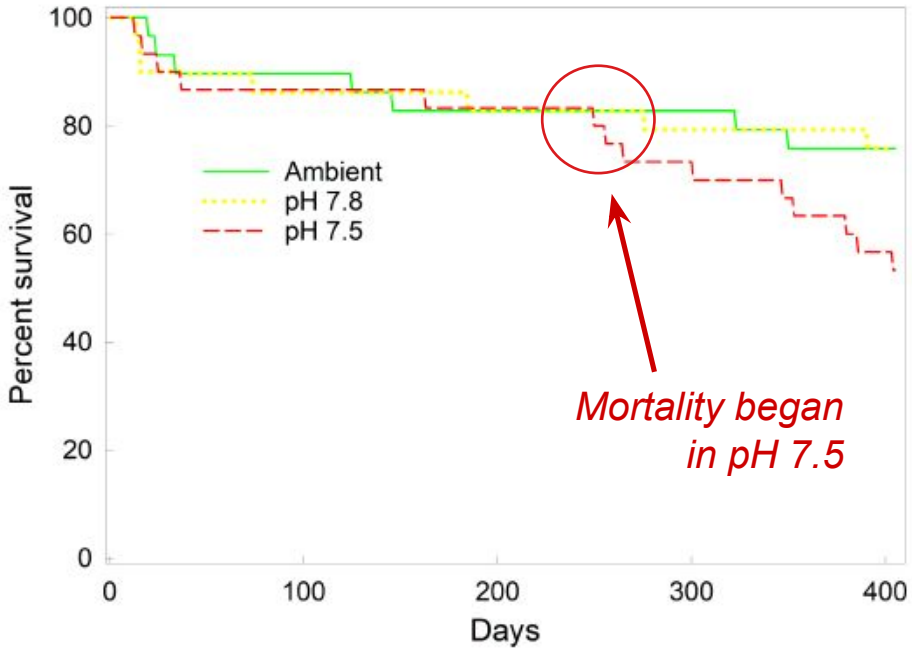
Morphometrics NMDS



No difference in growth, morphometrics over 396-day exposure

Also no effect on carapace width

Snow crab juveniles are surprising tolerant of long-term OA

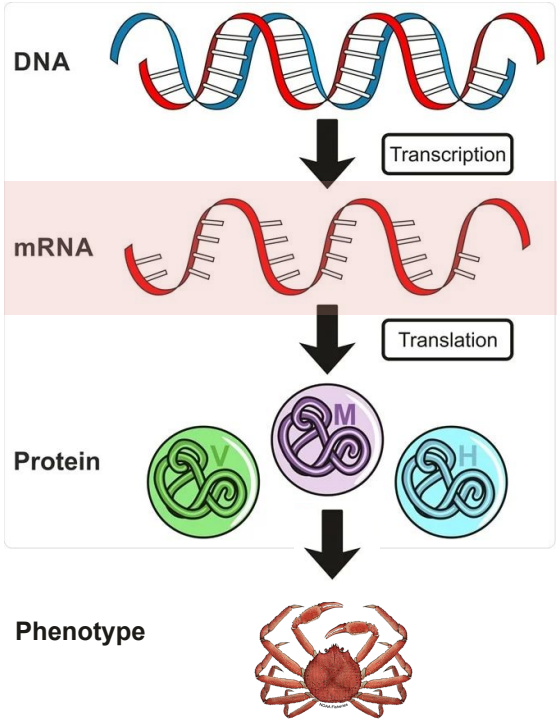


How do snow crab tolerate OA?



Survival unaffected until day 250

Gene expression analysis used to understand OA tolerance



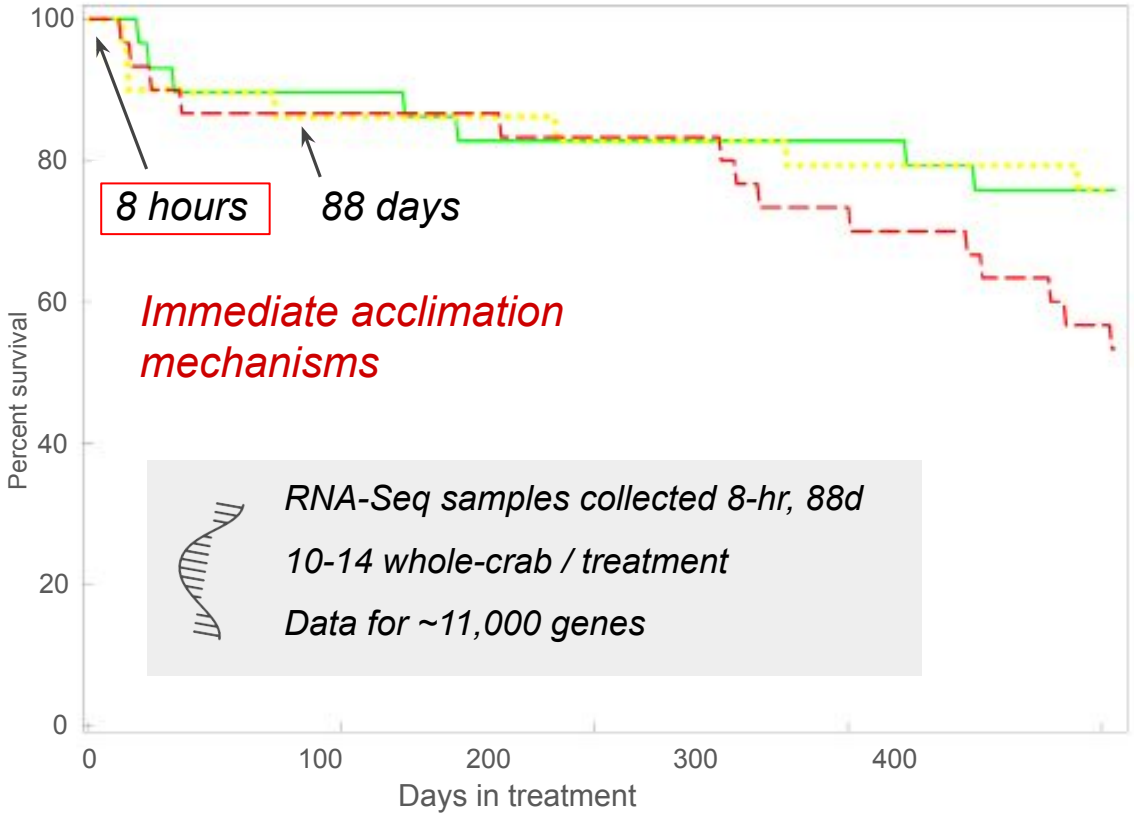
Adapted from: udaix/Shutterstock.com

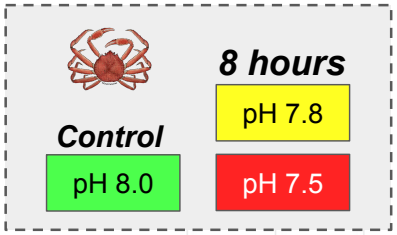
RNA-Seq measures all mRNA transcripts, ie. expressed genes

Identify genes, processes important to responding to a stressor

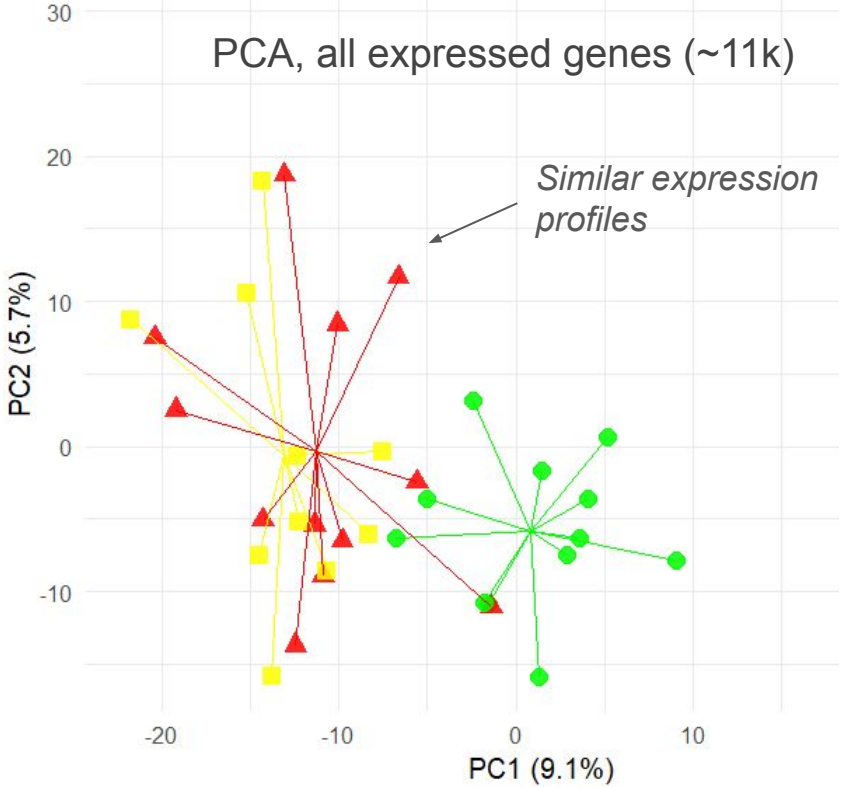
Captures shifts in energy allocation

Gene expression analysis used to understand OA tolerance





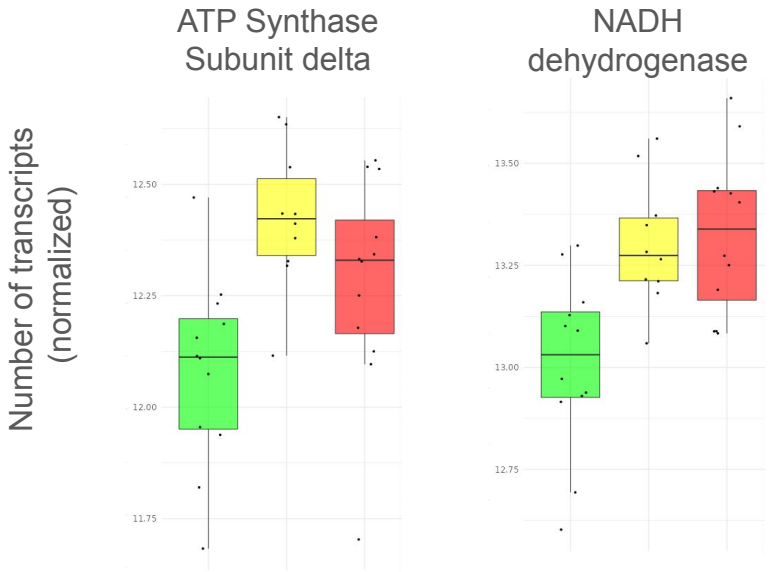
Immediate OA acclimation mechanisms

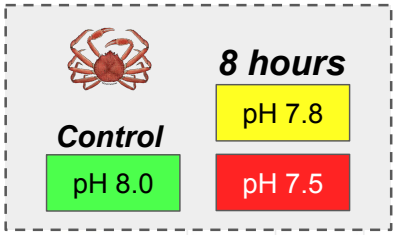


Mitochondrial genes highly active!

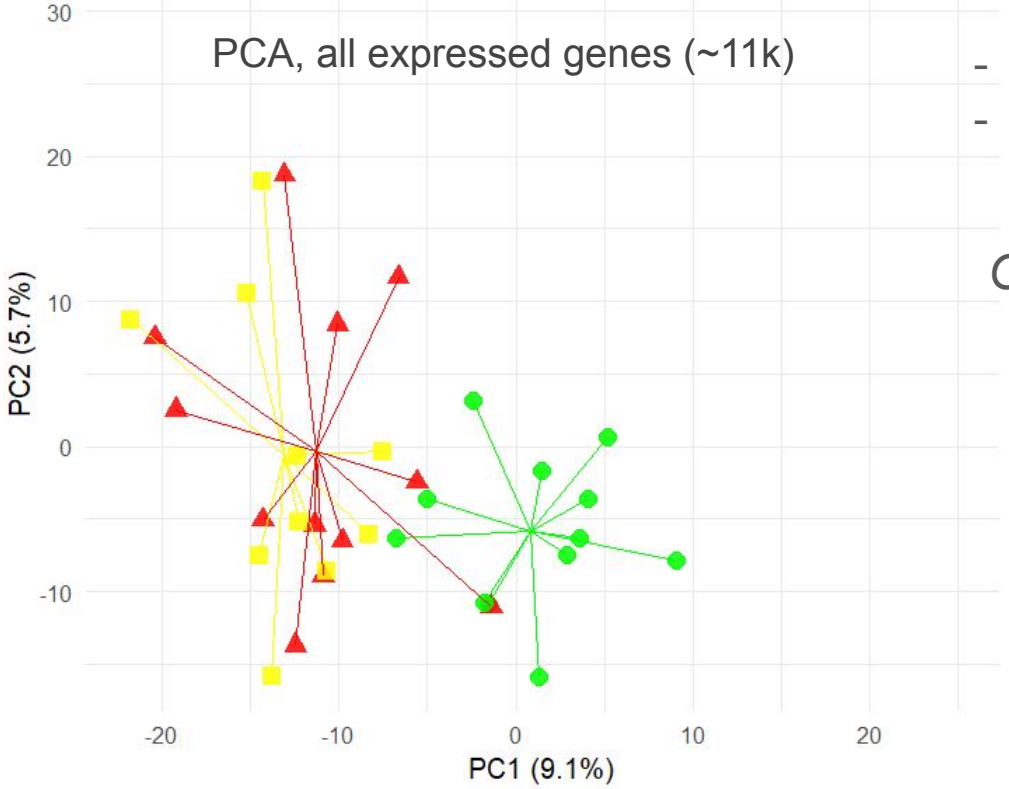
- Energy production
- Damage mitigation







Immediate OA acclimation mechanisms



Mitochondrial genes highly active!

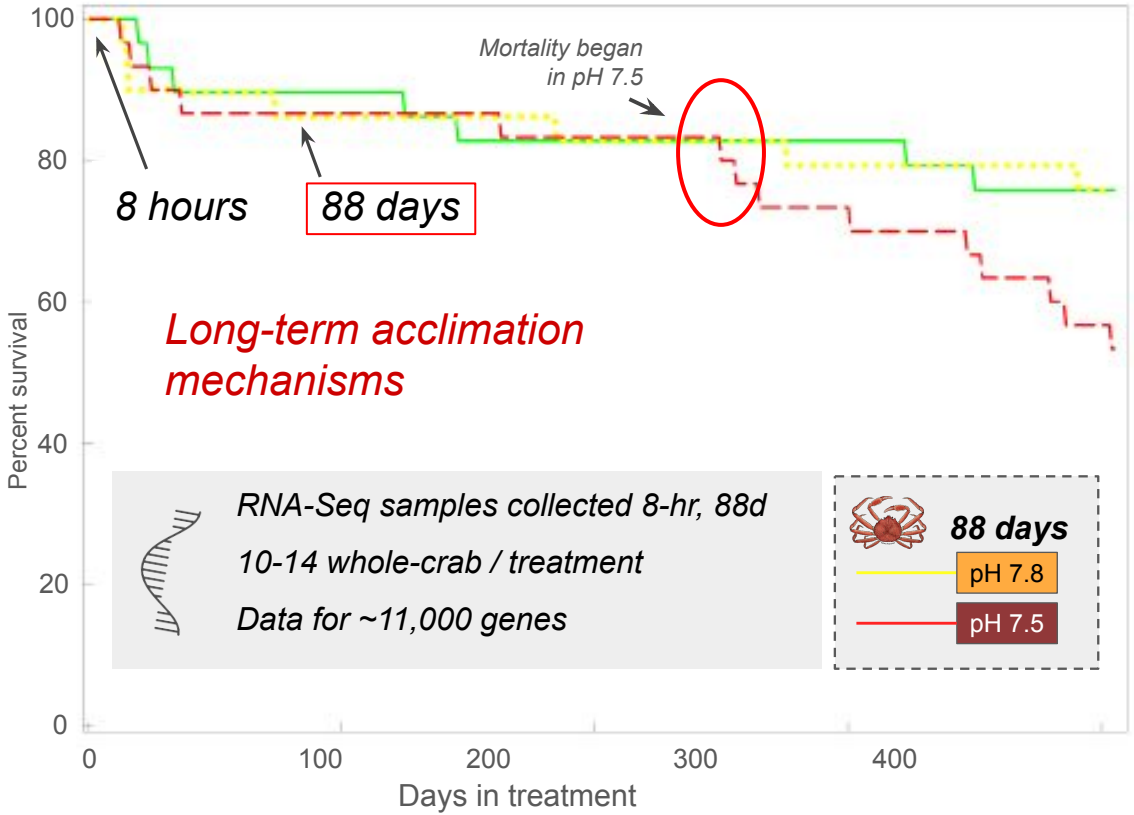
- Energy production
- Damage mitigation

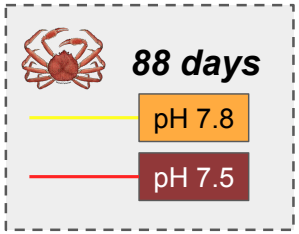


Other affected genes involved in…

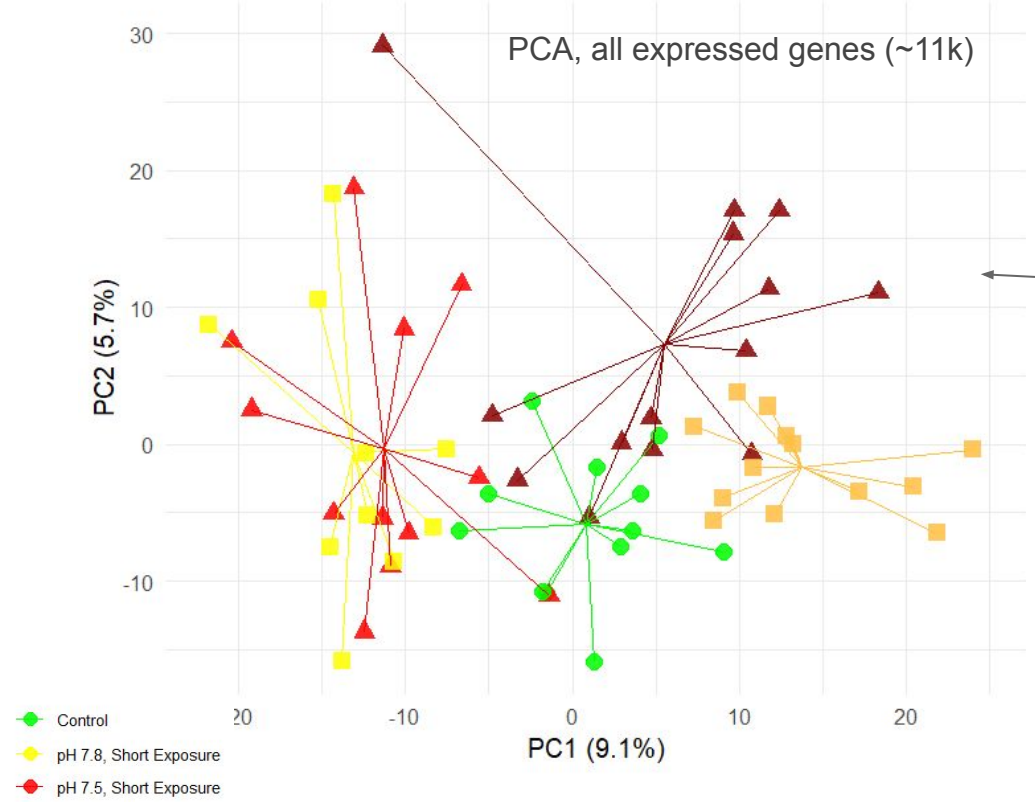
- Beta-oxidation
- Carbohydrate metabolism
- Protein processing
- Transcriptional regulation
- Intracellular transport

Gene expression analysis used to understand OA tolerance

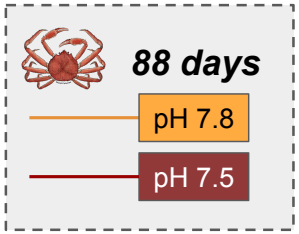




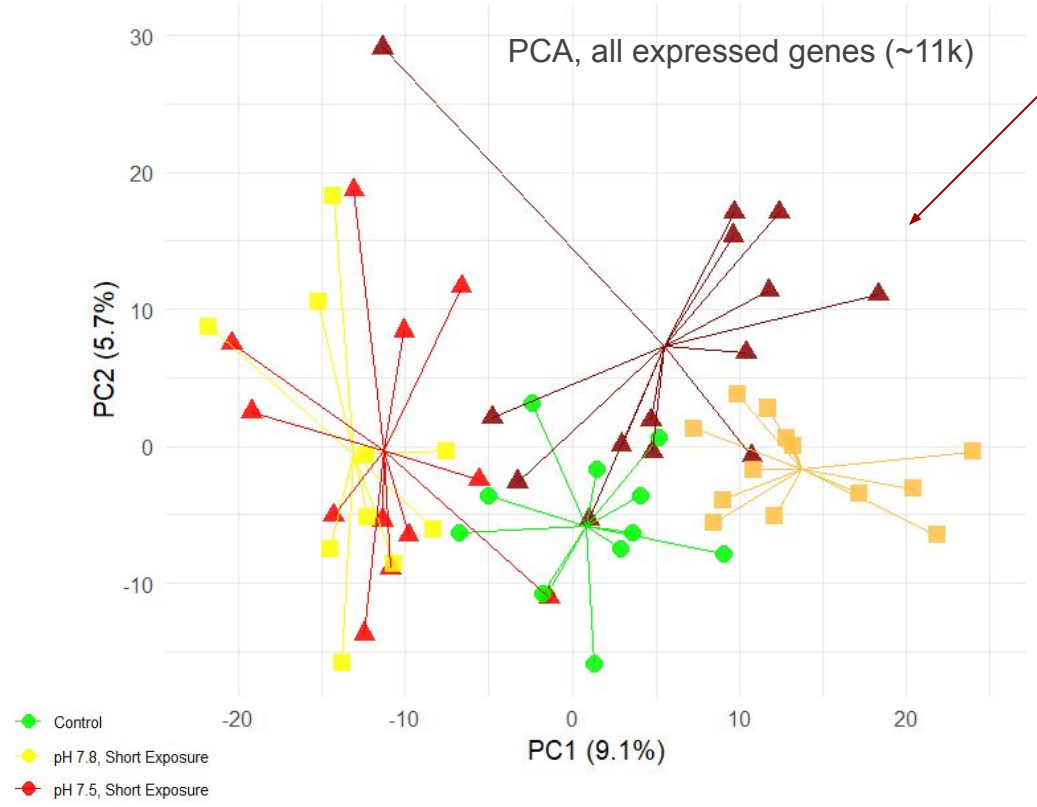
Chronic OA stress tolerance mechanisms



Compare expression of OA treatments at 88 days

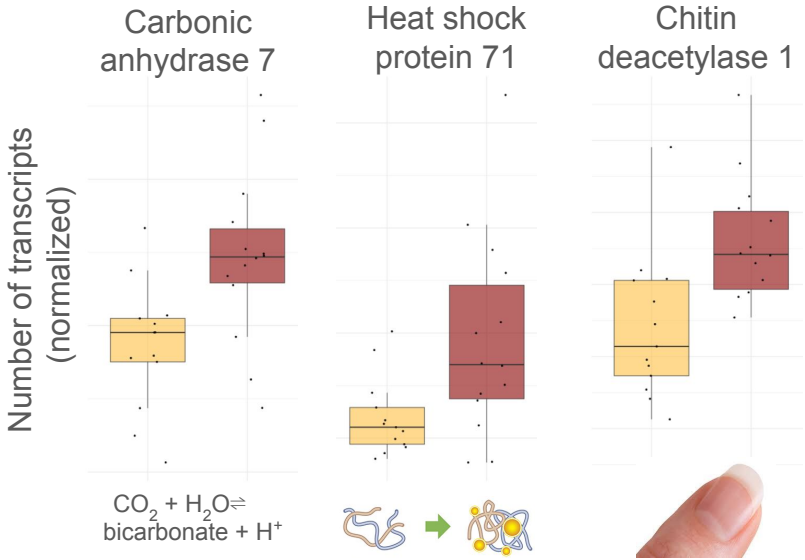


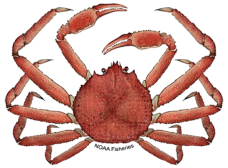
Chronic OA stress tolerance mechanisms



Higher activity, OA-stressed (pH 7.5)

- pH regulation
- Protein & DNA damage repair
- Chitin formation, modification





Short-term mechanisms, long term consequences

Snow crab juveniles are unaffected by moderate OA (pH 7.8), but more severe OA (pH 7.5) reduces survival, beginning at 250 days exposure.

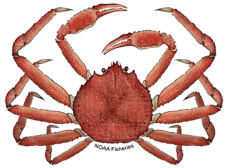
Short-term OA tolerance mechanisms …

- Investment in energy production, repair in mitochondria

Long-term OA tolerance consequences …

- Chronic cellular stress mitigation (protein, DNA damage) & chitin maintenance
- Mortality during molting - over-extended stress/damage-response systems

Snow crab OA gene expression explorer



Future work

Explore chronic OA stress biomarkers

- Heat shock proteins - responsive but universal to many stressors
- Carbonic anhydrase - top candidate from this study

Molecular comparisons to more OA-sensitive Tanner crab

- Tanners and snow crab hybridize, yet Tanners are highly OA-sensitive
- Tanner / snow gene expression comparison
- Tanner OA selection experiment

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Questions?